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Phylogenomic Challenges in Polyploid-Rich Lineages: Insights from Paralog Processing and Reticulation Methods Using the Complex Genus *Packera* (Asteraceae: Senecioneae)

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Abstract.—Phylogenomic discordance is pervasive and cannot always be resolved by increasing the amount of sequencing data alone. Biological processes such as polyploidy, hybridization, and incomplete lineage sorting are major contributors to discordance and must be accounted for to avoid misleading evolutionary interpretations. To better understand how these processes influence phylogenetic reconstruction, we conducted a comprehensive phylogenomic study in the complex genus *Packera*. With over 90 species and varieties, 40% of which exhibit polyploidy, aneuploidy, or other cytological complexities, *Packera* presents significant challenges for phylogenetic reconstruction. Given these complexities, we assessed different published paralog processing methods on the resulting evolutionary relationships and phylogenetic support of this group. We then applied three of these methods to evaluate their impact on tree topology and our understanding of *Packera*'s evolutionary history by constructing a time-calibrated phylogeny, reconstructing historical biogeography, and testing for ancient reticulation. Phylogenetic outcomes varied based on the paralog processing method used, with no method outperforming others. Our findings highlight the large impact of orthology inference and paralog processing on phylogenomic analyses, particularly in polyploid-rich groups such as *Packera*, and we offer guidance on methodological impacts along with practical recommendations. We note that gaining a robust understanding of *Packera*'s evolutionary history requires more than computational approaches alone. Although technological advancements have greatly expanded our ability to analyze genomic data, effective phylogenomic research still relies on strong taxon sampling and detailed species knowledge. Without careful attention to biological context, phylogenomic studies risk misinterpreting evolutionary history and processes. By integrating genomic results with knowledge of the study system, we can begin to improve the accuracy of evolutionary reconstructions and gain deeper insights into the complex history of plant diversification. [Compositae; discordance; hybridization; molecular systematics; orthology inference.]

The field of phylogenetics is drastically changing and now routinely incorporates multi-locus and genomic data. Although this expansion has enabled deeper insights into evolutionary relationships across the tree of life, it has also revealed persistent and complex challenges in phylogenetic inference. One significant challenge is gene tree discordance, where phylogenetic trees estimated from individual loci disagree with each other and/or with the inferred species tree (Degnan and Rosenberg 2009; Mendes et al. 2016). These incongruences are not simply a consequence of limited data; rather, they often reflect differences driven by underlying biological processes such as incomplete lineage sorting (ILS), gene duplication and loss, and hybridization or introgression (Pease et al. 2018). Such processes are especially prevalent in plants, where whole-genome duplications (WGDs), ancient hybridization, and other reticulate evolutionary processes are widespread and recurrent (Funk 1985; Vision et al. 2000; Wendel and Cronn 2003; Peirson et al. 2012; Panchy et al. 2016; Rose et al. 2021).

Polyploidy, often resulting from WGDs, is a dominant evolutionary force in angiosperms (Grant 1971; Soltis and Soltis 1999; One Thousand Plant Transcriptomes Initiative 2019), with some estimates suggesting that all extant flowering plants exhibit histories of polyploidy (Soltis and Burleigh 2009; Scarpino et al. 2014). These events gave rise to extensive paralogy in plant genomes, which poses significant challenges for orthology inference and phylogenetic analysis. Although these issues are particularly pronounced in plants (e.g., Asparagaceae, Bentz and Leebens-Mack 2024; Rosaceae, Morales-Briones et al. 2018, Morales-Briones et al. 2021a; and Amaranthaceae, Morales-Briones et al. 2021b), they are by no means exclusive to them. Similar complications have been documented in non-plants, such as African clawed frogs (Evans et al. 2004), desert rodents (Gallardo et al. 2004), alpine butterflies (Nice et al. 2013), and fish (Braasch and Postlethwait 2012), indicating that complex histories of gene duplication and horizontal gene transfer create phylogenetic uncertainty across the tree of life. As the scale and scope of phylogenomic data sets expand, so too does the need to carefully model and interpret the biological processes underlying gene tree discordance. Fortunately, the growing availability of phylogenomic data now makes it possible to start untan-

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gling these complex histories in ways that were not previously feasible; however, doing so requires tools and models that explicitly account for these processes.

In response, several bioinformatic tools and pipelines have been developed to infer orthology under increasingly realistic evolutionary models. Broadly, these approaches differ in how they treat gene duplication. Some pipelines retain a single representative gene copy per taxon, implicitly assuming that orthologous sequences can be identified or that one copy adequately reflects species history (e.g., HybPiper; Johnson et al. 2016). Others attempt to eliminate paralogy by excluding loci that exhibit evidence of duplication, thereby restricting analyses to putatively single-copy genes (e.g., Phyluce; Faircloth 2016). More recently, graph- and tree-based pipelines (e.g., paragone-nf; Yang and Smith 2014; Jackson et al. 2023) and species tree methods (e.g., ASTRAL-Pro; Zhang et al. 2020; Zhang and Mirarab 2022a) incorporate duplication and loss directly into inference, allowing multiple gene copies per taxon to contribute to species tree estimation. These strategies differ in their assumptions about the prevalence and identifiability of paralogy and therefore may yield markedly different phylogenetic outcomes, particularly in polyploid-rich systems where paralogs may be the rule rather than the exception. Yet, few studies have systematically compared how these alternative assumptions about paralogy influence both species tree inference and downstream evolutionary inferences.

This is especially relevant for groups such as *Packera* Á. Löve & D. Löve (Asteraceae: Senecioneae), a North American genus of about 91 species known for polyploidy, complex patterns of hybridization, and poorly resolved taxonomy. An estimated 40% of *Packera* taxa present polyploidy, aneuploidy, and/or other cytological disturbances (Barkley 1988; Trock 2006), complicating phylogenetic reconstruction of this group (Bain and Golden 2000). Given this, *Packera* provides an ideal system for testing how alternative strategies for han dling paralogs influence phylogenomic inference, owing to its extensive polyploidy and history of reticulation—all factors that increase the potential for gene tree discordance.

Here, we present the most comprehensive phylogenomic investigation of *Packera* to date, sampling nearly 93% of the genus and leveraging a high-resolution Hyb-Seq data set. To better understand how paralogy influences phylogenomic inference in duplication-rich lineages, we applied a robust comparative framework that evaluates multiple paralog processing strategies, including pipelines that remove all paralogs, those that model duplication and loss, and those that retain a single representative gene copy. We assessed how these approaches influence gene tree concordance, species tree topology, and node support. We then tested how three of these methods (a standard paralog processing method, a method that removes all paralogs, and a selective paralog-aware method; see the *Materials and Methods* section below) alter the interpretations of downstream macroevolutionary analyses including divergence dating, historical biogeography, and detection of ancient introgression. Our results demonstrate that the choice of paralog processing approach strongly influences inferred species relationships, gene tree concordance, node support, and evolutionary inferences, highlighting the profound effect that paralog-aware decisions can have in phylogenomics. Additionally, this study provides a broadly applicable framework for assessing paralogs and their downstream impacts on phylogenetic inference in systems shaped by gene duplication, reticulation, and rapid diversification.

Materials and Methods

Specimen Collection

A nearly complete sampling of currently recognized *Packera* species and varieties were used in this study (62 *Packera* species, 22 varieties, and 1 nothospecies; totaling 85 *Packera* taxa). *Packera* species were selected based on information published in the International Plant Names Index (IPNI; available at http://www.ipni.org) and then curated from discussions with botanists that have previously specialized in *Packera* (D. Trock & J. Bain, pers. comm.). Outgroup taxa from tribes Senecioneae ($n = 19$) and Anthemideae ($n = 4$) were selected by consulting recent studies of the group (Bain and Golden 2000; Pelser et al. 2010; Mandel et al. 2019). Additionally, three taxa from tribe Calenduleae were included for downstream analyses that require more distantly related outgroups that do not have a known WGD event (see "*1-to-1, MI, MO, and RT methods*" below; Barker et al. 2016; Huang et al. 2016; Zhang et al. 2021). A complete list of the 111 sampled species, herbarium vouchers, and their publication status can be found in Supplementary Table S1 (see 10.5061/dryad.bg79cnppn).

DNA Extraction and Sequencing

Leaf tissues were obtained from herbarium specimens or silica-dried field collections. Dried leaf tissue (~25 mg) was ground using a Bead Mill 24 Homogenizer (ThermoFisher Scientific, Atlanta, GA, USA), and total DNA was extracted using the E.Z.N.A. SP Plant DNA Kit (Omega Bio-tek, Atlanta, GA, USA) following the manufacturer's protocols with the addition of polyvinylpyrrolidone (PVP) and ascorbic acid to the first extraction buffer (10 mL SQ1 buffer, 100 mg PVP, 90 mg ascorbic acid per reaction). Samples of *Packera* and outgroup species that were obtained from other researchers followed different DNA extraction protocols, which are described in Supplementary Table S1. All DNA samples used in this study were quality checked using a Nanodrop 2000 (ThermoFisher Scientific, Carlsbad, CA, USA) and quantified using a Qubit High Sensitivity (HS) assay (ThermoFisher Scientific, Eugene, OR, USA). If the

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DNA had low concentrations or possible contaminants, the samples were cleaned following the E.Z.N.A. Cycle Pure Kit (Omega Bio-tek, Atlanta, GA, USA) and re-quantified with the Qubit HS assay. Up to 1 μg DNA per sample was sonicated with a QSonica machine (ThermoCube, Melville, NY, USA) to generate fragment sizes within the 400–500 base pair (bp) range. The amount of DNA used was dependent on the initial total DNA concentration of the targeted sample. Time allotted for sonication of each sample varied depending on approximate length and degree of fragmentation of the DNA. Prior to sonication, total DNA was visualized on a 1% agarose gel in 1X Tris-borate-EDTA (TBE) and GelRed 3x (Biotium, Fremont, CA, USA) and the approximate length was estimated by comparing with a 100 bp GeneRuler DNA ladder (ThermoFisher Scientific, Vilnius, Lithuania). If the DNA sample appeared smaller than 400 bp, it was not sonicated.

Libraries for each sample were prepared using the NEBNext Ultra II DNA Library Prep and NEBNext Multiplex Oligos for Illumina Kits (New England Biolabs, Ipswich, MA, USA) following the NEBNext Ultra II Version 5 protocol with size selection on DNA fragments at 300–400 bp range. The protocol was adjusted by halving all amounts of reagents and DNA used. The barcoded libraries were quality checked with an Agilent Bioanalyzer (Agilent Technologies, Santa Clara, CA, USA) and quantified by Qubit HS assay. Libraries were then used for targeted sequence capture with the custom MyBaits probe set, Compositae-1061 (Mandel et al. 2014), from Arbor Biosciences (Ann Arbor, MI, USA) following the manufacturer's protocol (v. 2.3.1). Approximately 500 ng of total DNA library in 6 μL of library, or equal amount of DNA per sample when pooling (maximum of eight samples), was used in each MyBaits reaction for hybridization with the baits for 36 h at 62 °C. Hybridized targets were recovered using Dynabeads M-280 Streptavidin (ThermoFisher Scientific, Vilnius, Lithuania) following the manufacturer's protocol. Captured targets were amplified and quantified using KAPA library quantification kits (Kapa Biosystems, Wilmington, MA, USA). A 1:4 ratio of library was added to each of the captured target pools to obtain off-target plastid sequence data. Libraries were sequenced on Illumina MiSeq (300 cycles) and NovaSeq 6000 (300 cycles) instruments at Psomagen (Rockville, MD, USA) across several runs. Sequence stats were obtained using the *pxlssq* function in Phyx v. 1.3.1 (Supplementary Table S2) (Brown et al. 2017).

Sequence Processing

Raw FASTQ data were trimmed of low quality bases and adapter contamination using Trimmomatic v. 0.36 (Bolger et al. 2014). The Sliding Window quality filter was used in this analysis (illuminaclip 2:30:10, leading 20, trailing 20, sliding window 5:20). Cleaned reads with a minimum length of 36 bp and corresponding forward and reverse pairs were retained. FastQC v. 0.11.9 (Andrews 2010) was used to ensure good sequence quality.

Plastid Assembly

Chloroplast sequences were extracted from trimmed reads using two approaches: the first to assemble complete plastomes from all samples with sufficient off-target sequencing data, and the second to create a multiple sequence alignment using available plastid data from all samples for phylogenetic analysis. First, the GetOrganelle v. 1.7.7.1 toolkit (Jin et al. 2020) was used to assemble circularized plastomes. The recommended parameters for embryophyte plant plastomes were used, except that the number of rounds and the disentangle time limit were increased (-R 75 and –disentangle-time-limit 7200). All complete circular plastomes were annotated using GeSeq (Tillich et al. 2017). Due to reports of conserved plastome structure in Senecioneae (Pascual-Díaz et al. 2021; Cho et al. 2024), a random *Packera* representative (*Packera cymbalaria* (Pursh) W.A.Weber & Á.Löve, see the *Results* section) with a complete plastome exhibiting the expected quadripartite structure was used as a reference for the second plastome assembly approach. In this, reads from each sample were mapped to the reference plastome using Bowtie2 (Langmead and Salzberg 2012), variants were called using SAMtools v. 1.18 and BCFtools v. 1.18 (Danecek et al. 2021), and consensus sequences were created using seqtk v. 1.4 (https://github.com/lh3/seqtk). For the aligned consensus sequences, model selection and maximum likelihood phylogenetic analysis were then conducted using IQ-TREE v. 2.2.2 (Minh et al. 2020) and visualized using "ggtree" (Yu et al. 2017) in R v. 4.3.1 (R Core Team 2016; RStudio 2020).

Nuclear Ortholog Assembly

Cleaned FASTQ data were assembled using the HybPiper v. 2.1.6 (Johnson et al. 2016) pipeline to match the reads to the target loci contained in the Compositae-1061 probe set (Mandel et al. 2014). In HybPiper, the command "assemble" was used to map the paired-end reads to the target loci using BWA v. 0.7.17 (Li and Durbin 2009). These were then assembled into contigs with specified kmer lengths (21, 33, 55, 77, and 99) using SPAdes v. 3.5 (Bankevich et al. 2012). A single representative sequence per taxon per locus was extracted using the "retrieve_sequences" command. Because HybPiper reconstructs target sequences by stitching together assembled contigs, these sequences may in some cases represent chimeric assemblies derived from multiple contigs and, in polyploid taxa, could potentially incorporate signals from different paralogous copies. Paralog detection was carried out using the "paralog_retriever" command (https://github.com/mossmatters/HybPiper/wiki/Paralogs), which flags

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loci that indicate potential paralogs, then extracts the assemblies that were flagged. A multifasta for each locus was generated and aligned in MAFFT v. 7.407 (Katoh and Standley 2013) using default parameters.

Phylogenetic Analyses

Given the complex history of hybridization and polyploidization events in *Packera*, along with the large number of paralogs flagged from HybPiper in the current study (Supplementary Table S3), we sought to compare how different published orthology inference methods and species tree estimation tools impacted the inferred evolutionary relationships and support of this group. We generated multiple nuclear phylogenies utilizing different paralog processing and species tree estimation methods using the single- and multi-copy orthologous sequences, or orthologs, designated by HybPiper: HybPiper with concatenation (hereafter "Concatenation"; individual gene histories of putative single-copy orthologs are not accounted for in model selection), HybPiper with ASTRAL-III (hereafter "HybPiper-df"; putative single-copy orthologs are chosen based on coverage depth and sequence similarity concerning the gene references from Compositae-1061, then the species tree was inferred using coalescent methods), HybPiper with weighted ASTRAL (hereafter "wASTRAL"; similar to HybPiper-df, but low support gene tree branches are contracted prior to species tree estimation), HybPiper with ASTRAL-Pro (hereafter "ASTRAL-Pro"; all single-copy and multi-copy genes identified by HybPiper are considered, and the species tree that optimizes a quartet similarity measure is chosen), and HybPiper with the "one-to-one orthologs" (1-to-1), "maximum inclusion" (MI), "monophyletic outgroup" (MO), and "rooted ingroup" (RT) paralog processing methods by Yang and Smith (2014), with species trees inferred using ASTRAL-III (further details below; refer to Supplementary Table S4 and Supplementary Fig. S1). By performing comparisons among methods, we aimed to gain a better understanding of the impact of paralogy on phylogenomic studies, and ultimately increase phylogenomic support and concordance across species of complex lineages.

Concatenation.—Aligned contigs of the putative single-copy orthologs designated by HybPiper were concatenated using FASconCAT-G v. 1.02 (Kück and Longo 2014), and then analyzed in PartitionFinder v. 2.1.1 (Lanfear et al. 2012) to determine the most appropriate nucleotide substitution model (GTR, GTR+Γ, and GTR+I+Γ). The overall best-fitting model (Supplementary Table S5) was used to build a concatenated nuclear maximum likelihood tree in RAxML v. 8.1.3 (Stamatakis 2014) with 1000 bootstrap (BS) replicates.

HybPiper-df and wASTRAL.—For the HybPiper-df and wASTRAL analyses, putative single-copy nuclear gene alignments were separated based on the most suitable nucleotide substitution model for each gene as determined by PartitionFinder (Supplementary Table S5), and then individual gene trees were generated using RAxML. For the HybPiper-df analysis, the gene trees were used in ASTRAL-III v. 5.7.3 (Zhang et al. 2018) to generate the species trees. Local posterior probability (LPP) values were estimated for branch support and indicate the probability that the branch is the true branch given the set of gene trees provided (Sayyari and Mirarab 2016).

For the wASTRAL analysis, a species tree was generated using the "astral-hybrid" weighting scheme implemented in the ASTER program (https://github.com/chaoszhang/ASTER). This method has shown to be more accurate than typical HybPiper-df methods because wASTRAL incorporates gene tree uncertainty by weighting quartets based on the confidence levels of the input gene trees. In doing so, wASTRAL reduces the influence of erroneous gene trees, leading to a more reliable species tree estimation. Support was weighted using approximate Bayesian (aBayes) supports, which range from 0 to 1 and have been shown to perform better than bootstrapping (Zhang and Mirarab 2022b).

ASTRAL-Pro.—ASTRAL for PaRalogs and Orthologs (ASTRAL-Pro2) v. 1.13.1.3 (Zhang et al. 2020; Zhang and Mirarab 2022a), hereafter referred to as ASTRAL-Pro, provides a quartet-based species-tree inference method that allows multi-copy gene trees to be used as input. Thus, all paralogous and orthologous sequences as assigned by HybPiper were used to generate the species tree instead of selecting a single ortholog to represent an individual gene's history (e.g., HybPiper-df method). By accommodating multi-copy gene trees, ASTRAL-Pro accounts for gene duplication and loss events, reducing biases that arise from treating paralogs as true orthologs. This approach is intended to improve species tree accuracy, particularly in data sets with extensive gene family expansions, by leveraging more information from complex gene histories. The ASTRAL-Pro pipeline is similar to HybPiper-df, as described above, but differs by having the user generate separate alignments and conduct maximum likelihood analyses on all orthologs, including all genes flagged as paralogous and single-copy identified by HybPiper. Single-copy and paralogous gene sequences were independently aligned using MAFFT, and gene trees were generated under the best-fitting nucleotide substitution model from PartitionFinder (Supplementary Table S5) in RAxML with 1000 BS replicates. The RAxML output files were then used as input for ASTRAL-Pro2 to generate the species tree with LPP values at each node.

1-to-1, MI, MO, and RT methods.—Paralog processing was carried out using the tree-based orthology inference strategies originally proposed by Yang and Smith (2014) via the paragone-nf v. 1.1.0 pipeline (Jackson et al. 2023; https://github.com/chrisjackson-pellicle/paragone-nf). This pipeline investigates the loci flagged as paralogous from HybPiper and determines which loci can be retained or pruned given specified conditions

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(Yang and Smith 2014; Morales-Briones, Gehrke, et al. 2021). Briefly, gene files from the "paralog_retriever" command, which includes all orthologs and paralogs flagged by HybPiper, were aligned using MAFFT with default settings, and then gene trees were generated using IQ-TREE. Only tips with the highest number of unambiguous characters were retained for each gene, whereas mono- and paraphyletic tips belonging to the same taxon were masked. Rogue tips were removed with TreeShrink v. 1.3.2 (Mai and Mirarab 2018) using default settings. From there, the fasta sequences corresponding to the trimmed tips are removed from the alignment files generated by MAFFT. Outgroup taxa without a known WGD event (here tribe Calenduleae: *Calendula arvensis L.*, *Tripteris pinnatilobatum* (Norl.) B.Nord., and *Chyrsanthemoides incana* (Burm.f.) Norl.) are then added to the fasta files and realigned with MAFFT and new gene trees are made. These new genes trees are then applied to the four orthology inference methods proposed by Yang and Smith (2014)—the "one-to-one orthologs" (1-to-1), "maximum inclusion" (MI), "monophyletic outgroup" (MO), and "rooted ingroup" (RT) methods. Briefly, the 1-to-1 approach is the simplest strategy in that it only retains homologs that have no identified duplications. The MI approach iteratively extracts the subtrees with the highest number of taxa without taxon duplications. The MO approach filters for homolog trees where outgroup taxa are monophyletic and non-repeating. After filtering, ingroup paralogs are pruned to retain subtrees that maximize taxon sampling while maintaining single-copy orthology. The RT approach searches subtrees of ingroup taxa, roots them by the designated outgroups, and then sequentially prunes inferred duplications from the root toward the tips, retaining one copy per taxon in each subtree (Yang and Smith 2014; Morales-Briones, Gehrke, et al. 2021).

To aid in consistency across analyses, the HybPiper run described above was used as input instead of using the hybpiper-nf pipeline (Jackson et al. 2023). The three Calenduleae taxa were considered the internal outgroups because a recent WGD event has not been reported for this tribe, as is required for the MI and MO analyses (Yang and Smith 2014). The resulting trimmed alignments produced via paragone-nf were used as input for RAxML to generate gene trees, and a species tree with LPP support values was generated using ASTRAL-III following the "HybPiper-df" protocol described above.

Measuring Phylogenetic Tree Support

The normalized quartet score was calculated for all orthology inference methods that generated a species tree with ASTRAL-III using the "-q" option. Branch lengths were recalculated from coalescent units to nucleotide substitutions per codon site for each ASTRAL-III species tree with IQ-TREE, which uses the supermatrix and individual species tree as input. Parsimony informative sites (PIS) and pairwise identity (PI) were calculated for each gene matrix using the *parsimony_informative_sites* and *pairwise_identity* functions, respectively, in phykit v. 2.0.1 (Steenwyk et al. 2021). PIS quantifies the number and proportion of sites that contribute to phylogenetic signal, whereas PI reflects the average sequence similarity among taxa and can serve as a proxy for relative evolutionary rate (Steenwyk et al. 2021). Nuclear and plastome trees were visualized using the package "phytools" (Revell 2012) in R v. 4.0.5.

Measuring Discordance

Species trees from concatenation, plastid, HybPiper-df, wASTRAL, and paragone-nf (1-to-1, MI, MO, and RT) were analyzed for underlying discordance using Quartet Sampling (Pease et al. 2018). ASTRAL-Pro was not analyzed as Quartet Sampling does not allow multi-copy gene trees as input. Quartet Sampling assesses the confidence, consistency, and informativeness of internal tree relationships, and the reliability of each terminal branch by evaluating if a phylogeny has a lack of support due to lack of information, discordance from ILS or introgression, or misplaced or erroneous taxa. Three scores were calculated for each internal branch of the focal tree, which determines the consistency of information (Quartet Concordance [QC]), the presence of secondary evolutionary histories (Quartet Differential [QD]), and the amount of information (Quartet Informativeness [QI]), indicated as QD/QC/QI at each node. Quartet Fidelity (QF) scores were calculated at each tip to report how frequently a taxon is included in concordant topologies. Using the rooted species trees and concatenated gene alignments from FASconCAT-G (described above), Quartet Sampling was run using default settings on the designated analyses. The concatenation, plastid, HybPiper-df, wASTRAL, 1-to-1, MI, and MO species trees were rooted to the outgroup Calenduleae taxon, *Tripteris pinnatilobatum*, whereas the RT tree was rooted to *Doronicum pardalianches* L., because the Calenduleae taxa were removed through the paragone-nf analysis, using the *pxrr* function in Phyx (Supplementary Table S4).

Additionally, PhyParts v. 0.0.1 (Smith et al. 2015) was used with default settings to quantify and visualize discordance in the HybPiper-df, wASTRAL, and paragone-nf (1-to-1, MI, MO, and RT) trees. Similar to Quartet Sampling, ASTRAL-Pro could not be analyzed because of multi-copy gene trees. Additionally, the concatenation and plastid trees could not be analyzed because those analyses do not use individual gene histories as input. Resulting trees containing pie charts indicating whether gene trees were supported, conflicted, uninformative, or support the top alternative bipartition at each node were generated using a custom python script (https://github.com/mossmatters/phyloscripts/tree/master/phypartspiecharts). Phy-

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Parts requires rooted gene trees and a rooted species tree as input. Similar to Quartet Sampling, HybPiper-df, 1-to-1, MI, and MO gene trees were rooted to *Tripteris pinnatilobatum* and the RT tree was rooted to *Doronicum pardalianches* using the *pxrr* function in Phyx. Rooting gene trees for PhyParts decreased the number of genes used as input because not all gene trees contained the targeted outgroup taxon (see Supplementary Table S4). For phylogenies generated with ASTRAL technologies—HybPiper-df, wASTRAL, ASTRAL-Pro, and paragone-nf trees (Supplementary Table S4)—introgression and ILS were visualized across lineages using phytop v. 0.3.1 (Shang et al. 2025) with default settings. The species trees were re-ran with the flag "-t 2" added to be used as input for phytop.

Tree Topology Comparison

To evaluate similarity among the nuclear gene trees, pairwise adjusted Robinson–Foulds (RF_{adj}) distances were calculated as outlined by Mitchell et al. (2017) between all trees using the *RF.dist()* function in package "phangorn" (Schliep 2011) in R, with unrooted trees as input and setting the "normalize" argument to TRUE (Steel and Penny 1993). The Calenduleae outgroup taxa were trimmed from the trees in R using the *drop.tip()* function in package "ape" (Paradis and Schliep 2019) to make concatenation, HybPiper-df, ASTRAL-Pro, and paragone-nf trees directly comparable. The RF_{adj} values range from zero to one, indicating whether the tree topologies are identical or completely dissimilar, respectively. A principal component analysis (PCA) was generated from all RF_{adj} values in R to aid in visualization.

To assess the robustness of inferred relationships across phylogenetic inference methods, we quantified topological congruence among species trees inferred using ASTRAL-based approaches (Supplementary Table S4). For each tree, we pruned all outgroup taxa and collapsed nodes with low support (LPP < 0.9) using base R to avoid counting weakly supported relationships as evidence of concordance. We then constructed a majority-rule consensus tree ($P = 0.5$) across the support-filtered trees using "ape," and extracted bipartitions to determine which methods recovered each consensus clade. Node labels therefore indicate the number of methods and the specific analytical frameworks supporting each relationship.

All nuclear and plastid phylogenies and their associated analyses described above were compared, but for simplicity only the HybPiper-df, 1-to-1, and MO trees were analyzed in further analyses. HybPiper-df was chosen as it selects the putative single-copy orthologs less rigorously than the remaining methods, creating a baseline method for comparison. We then compared HybPiper-df to a method that disregards all genes containing paralogs (1-to-1) and a more rigorous paralog processing method (MO). We chose MO over MI and RT due to its more stringent ortholog identification, which retains only loci with monophyletic outgroups, thereby minimizing the inclusion of paralogs, reducing phylogenetic noise, and providing a more reliable foundation for downstream phylogenetic analyses (Yang and Smith 2014; Morales-Briones, Gehrke, et al. 2021; Jackson et al. 2023).

Molecular Clock Dating

Estimates of divergence times were calculated on the HybPiper-df, 1-to-1, and MO trees with RelTime (Tamura et al. 2012, 2018) using the command-line version of the MEGA software, MEGA-CC v. 11.0.11 (Kumar et al. 2012, 2016), with necessary input files generated with MEGA-X (Kumar et al. 2018; Mello 2018). The three Calenduleae taxa were used as outgroup taxa for this analysis.

There are no fossils available within *Packera* or tribe Senecioneae, so instead, molecular clock dating utilized both primary and secondary age constraints (Supplementary Table S6). Briefly, a maximum (59.95 myr) and minimum (31 myr) age was tested for the Anthemideae/Senecioneae split given those dates are the highest and lowest values within the 95% CI from a previous diversification rate analysis of Asteraceae (Mandel et al. 2019). Next, a maximum age of 34.06 myr was set at the *Packera* crown node from the upper age limit provided by Mandel et al. (2019). Additionally, a minimum age constraint of 12 myr was tested within the outgroup (the Anthemideae clade) given the presence of a fossil pollen (Wang 2004; Hobbs and Baldwin 2013; Palazzesi et al. 2022). Finally, a maximum age constraint of 14.3 myr was tested for *Pericallis* D.Don (Senecioneae), a genus endemic to the Macaronesia islands (Jones et al. 2014), because that is the estimated age of Porto Santo, the oldest volcanic archipelago within the Madeira Islands (McDougall and Schmincke 1976; Geldmacher et al. 2000).

Multiple dating scenarios were tested using the primary and secondary age constraints listed above (Supplementary Table S6): Scenario 1 included all primary and secondary calibrations. Scenarios 2 and 3 excluded the maximum time constraint for *Packera* and the Senecioneae/Anthemideae split, respectively, whereas scenario 4 excluded both. Scenarios 5 and 6 both excluded the minimum and maximum time constraints of the Senecioneae/Anthemideae split, but scenario 6 also did not include the *Packera* maximum (Supplementary Table S6). Results of all tested scenarios were compared for biological relevance, though only one scenario was chosen for the discussion and ancestral biogeography analysis (see below). This approach allowed us to focus on the topology that best aligned with prior systematic understanding and showed the strongest overall support across analytical frameworks.

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Biogeography

Ancestral ranges were estimated on the chosen dated phylogenies of HybPiper-df, 1-to-1, and MO using the package BioGeoBEARS v. 1.1 (Matzke 2014) in R. BioGeoBEARS estimates historical, biogeographic ranges by implementing popular models in a likelihood framework. Some of the most common models used are dispersal-extinction-cladogenesis (DEC; Ree and Smith 2008), the likelihood version of DIVA (DIVALIKE; Ronquist 1997), and the BayArea likelihood version of the range evolution model (BAYAREALIKE; Landis et al. 2013). In BioGeoBEARS, founder-event speciation can be added to any of these models (+*J*) with flexibility in determining relative probability. Likelihood Ratio Test (LRT) and Akaike Information Criterion (AIC) were used to determine which model best fits the data.

All *Packera* taxa, along with their four most closely related outgroup genera (*Pericallis*, *Elekmania* B.Nord., *Werneria* Kunth, and *Xenophyllum* V.A.Funk), were coded as present or absent in 10 geographic ranges outlined in Takhtajan (1986) (Supplementary Table S7): Arctic Province, Canadian Province, Appalachian Province, Atlantic and Gulf Coast Plain Province, North American Prairies Province, Rocky Mountain and Vancouverian Province, Madrean Region, Asia, Europe, and Central America. Distributions were determined from previously published literature (Freeman and Barkley 1995; Trock 1999, 2006; Gramling 2006; Mahoney and Kowal 2008; Elven et al. 2011; Yeatts et al. 2011). Only 4 of the 26 outgroup taxa were included to limit the number of geographic ranges used in the analysis. The remaining outgroup taxa were pruned from the dated tree generated with RelTime using the *pxrmt* function in Phyx. The maximum range size was constrained to four areas to reflect the empirical distribution of *Packera*, in which the vast majority of taxa occupy one to four regions.

Reticulation

The Species Networks applying Quartets (SNaQ) function in the Julia v. 1.8.1 (Bezanson et al. 2018) package PhyloNetworks v. 0.15.2 (Solís-Lemus and Ané 2016; Solís-Lemus et al. 2017) was used to investigate reticulate evolution among *Packera* taxa. This package uses unrooted gene trees to infer phylogenetic networks in a pseudolikelihood framework with multi-locus data, while accounting for ILS and reticulation. We tested the fit of multiple models and gradually increased the number of reticulation events (h) from 0 to 10 to find which pseudolikelihood score reached a plateau, given recommendations by Solís-Lemus et al. (2017). This process is computationally intense, so only clades containing 14 taxa or less that had counter-support nodes in the HybPiper-df, 1-to-1, and/or MO trees, as determined by a negative QC score resulting from Quartet Sampling, were used as input for PhyloNetworks, with the remaining taxa pruned from the tree using the *pxrmt* function in Phyx. Given this, five *Packera* clades were investigated for ancient reticulation events. Additionally, a single species from each major clade was then used to represent backbone relationships throughout the entire phylogeny, totaling six separate PhyloNetwork runs that will be referenced as follows: Backbone, representing 11 taxa that span the entire phylogeny; Eastern, a clade of 12, 13, or 15 species (MO, 1-to-1, and HybPiper-df, respectively) either distributed along eastern North America or having known affiliations with those species; California, eight taxa with distributions mainly within California; Mexico, seven (MO and HybPiper-df) to eight (1-to-1) taxa with distributions mainly within Mexico; Rocky, a clade of five taxa that are distributed in the Central USA, most within the Rocky Mountains; and Arctic, nine taxa that are present in colder climates along the Pacific Northwest and into Canada (Supplementary Table S8). Rocky could not be run for 1-to-1 because fewer than five taxa were present in that target clade.

Results

Sequencing and Matrix Results

Illumina sequencing resulted in approximately 733.3 million reads across 111 taxa. Read numbers ranged from 535,934 in *Werneria aretioides* Wedd. to 20,685,576 in *Senecio vulgaris* L. (full list in Supplementary Table S2). HybPiper retained 1058 of the targeted 1061 loci across all 111 taxa, with 846 of those loci having paralog warnings (~80%). The number of loci recovered for each taxon ranged from 226 in *Packera paupercula var. paupercula* (Michx.) Á.Löve & D.Löve to 1007 in *Tripteris pinnatilobatum* (Supplementary Table S3). The number of gene trees varied across the different paralog processing and phylogenetic tree inference analyses, from 285 in the 1-to-1 analysis to 7388 in the MI analysis (Supplementary Table S4). PIS also ranged from 2.11% in the plastid tree to 99.64% in the MO tree (Supplementary Table S4). For species trees generated with ASTRAL-III, normalized quartet scores were used to measure gene tree discordance, which ranged from 0.516 for the RT tree to 0.534 for the MO tree (Table 1), indicating that ~52–53% of the gene trees matched the species tree in those analyses.

Using GetOrganelle (Jin et al. 2020), off-target plastid sequencing resulted in 93 partial and 18 complete plastomes. Complete plastomes from 17 *Packera* and 1 *Senecio* species ranged from 150,772 to 151,728 bp in size (Supplementary Fig. S2). Of these, the plastome from *Packera cymbalaria* was used as the reference assembly for mapping reads from all other samples. For each mapped consensus sequence, the percentage of missing data varied drastically, with *Tephroseris newcombei* (Greene) B.Nord. & Pelser exhibiting the most missing data (93.1%; Supplementary Table S9 and Supplementary Fig. S3). The multiple sequence alignment with 111 taxa had 2.11% PIS (Supplementary Table

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TABLE 1. Summary of support results across the nine nuclear and plastid phylogenetic trees generated with 111 taxa.

Analysis	Num. high support nodes (≥0.9 PP/90 BS)	Num. low support nodes (<0.9 PP/90 BS)	Normalized QS score	Num. full support QS nodes (1/−/1)	Num. counter-support QS nodes (QC < 0)	Num. nodes rejecting ILS ($P < 0.05$)	Max IH index
Plastid	51	57	N/A	0	28	N/A	N/A
Concatenated	48	60	N/A	11	25	N/A	N/A
HybPiper-df	59	49	0.517	11	28	17	0.62
wASTRAL	64	44	N/A	10	24	21	0.498
ASTRAL-Pro	49	59	N/A	N/A	N/A	10	0.483
MI	65	43	0.519	12	26	16	0.853
MO	50	58	0.534	9	23	9	0.51
RT	62	43	0.516	10	22	17	0.683
1to1	35	73	0.529	8	27	4	0.426

Note: PP = posterior probability; BS = bootstrap; QS = Quartet Sampling; QC = quartet concordance factor generated from Quartet Sampling; ILS = incomplete lineage sorting; IH = introgression/hybridization index as indicated by PhyTop.

S4). All tree and alignment files can be found online (see 10.5061/dryad.bg79cnppn).

Topological Comparisons and Discordance

Phylogenetic topologies varied depending on genomic source (e.g., nuclear or plastid) as well as paralog processing and phylogenetic tree inference method (i.e., HybPiper-df, ASTRAL-Pro, 1-to-1, etc.) (Figs. 1 and 2; Supplementary Fig. S4). When building phylogenies with nuclear data, *Packera* was generally placed as the sister lineage to a clade comprising *Pericallis* and *Elekmania*, with the exception of ASTRAL-Pro, which placed *Packera* as sister to *Tephroseris* and in a larger clade with *Pericallis* and *Elekmania* (Fig. 1). Interestingly, the plastid data showed a similar relationship to ASTRAL-Pro; however, *Pericallis* was placed unexpectedly in a separate clade with other Senecioninae taxa. All nuclear phylogenies recovered monophyletic tribes and subtribes (e.g., Senecioninae and Tussilaginineae s.str.), though the relationships within the subtribes varied. The mapped plastid tree generally separated the subtribes, apart from *Roldana* (Senecioninae) in Tussilaginineae s.str. (Fig. 1; Supplementary Figs. 3 and 4). The placement of *Tephroseris* varied across phylogenies, with its most consistent placement as sister to the rest of Senecioneae (Fig. 1). *Doronicum* was consistently placed sister to Senecioneae plus Anthemideae, providing support for its elevation to tribal status (Doroniceae; Panero 2005; Fu et al. 2016).

Phylogenetic topologies remained highly incongruent independent of methods used (Figs. 1 and 2; Supplementary Figs. 5 and 6 and Supplementary Table S10). The HybPiper-df and wASTRAL tree topologies were most similar ($RF_{adj} = 0.324$) (Fig. 1; Supplementary Table S10), whereas the plastid tree was consistently the most dissimilar to the remaining trees (RF_{adj} ranging from 0.895 to 0.914), with the concatenated tree being the next most dissimilar (RF_{adj} ranging from 0.562 to 0.648; Supplementary Table S10). Additionally, the normalized quartet scores from HybPiper-df indicate that only about half of the gene trees match the species trees in the HybPiper-df and paragone-nf analyses (Table 1). Most of the topological conflict was among *Packera* taxa, with minor differences among the remaining Senecioneae taxa and outgroups (Supplementary Fig. S4).

Given this extensive topological incongruence in *Packera*, we generated a majority-rule consensus phylogeny across all ASTRAL-based trees (Fig. 2) to evaluate which relationships were consistently recovered across analytical frameworks. No internal *Packera* relationships were recovered by all seven ASTRAL methods, and only a small subset of clades were supported by more than half of the approaches. Topological instability was most pronounced among larger clades (> 4 taxa), the majority of which were recovered in only one or two analyses. The most consistently recovered larger grouping was the Mexico clade, which was retained by three methods (HybPiper-df, MO, and RT). In contrast, smaller clades (≤ 4 taxa) showed comparatively greater agreement, with two to five methods supporting those relationships (Fig. 2). Overall, concordance among methods declined with increasing clade size, indicating that deeper relationships within *Packera* are particularly sensitive to analytical framework.

Support values varied depending on which genomic source and method was used, though generally remained low. The wASTRAL and MI trees showed the highest node support with 58 out of 110 nodes (52.7%) having ≥ 0.95 LPP (Table 1; Supplementary Fig. S4). In contrast, the plastid and concatenated trees showed much lower resolution with the fewest number of nodes having high support (36 of 110 nodes, 32.7%, have ≥ 95 BS; Table 1; Supplementary Fig. S4).

Quartet Sampling (QS), which evaluates support and conflict for branches using a quartet-based approach, indicated that underlying gene tree discordance remained high, independent of the method used. HybPiper-df and concatenated trees had the most fully supported nodes (when QS = 1/−/1) when compared with the remaining phylogenies (Supplementary Fig. S7), though the number was still notably low (~10%). Even so, the

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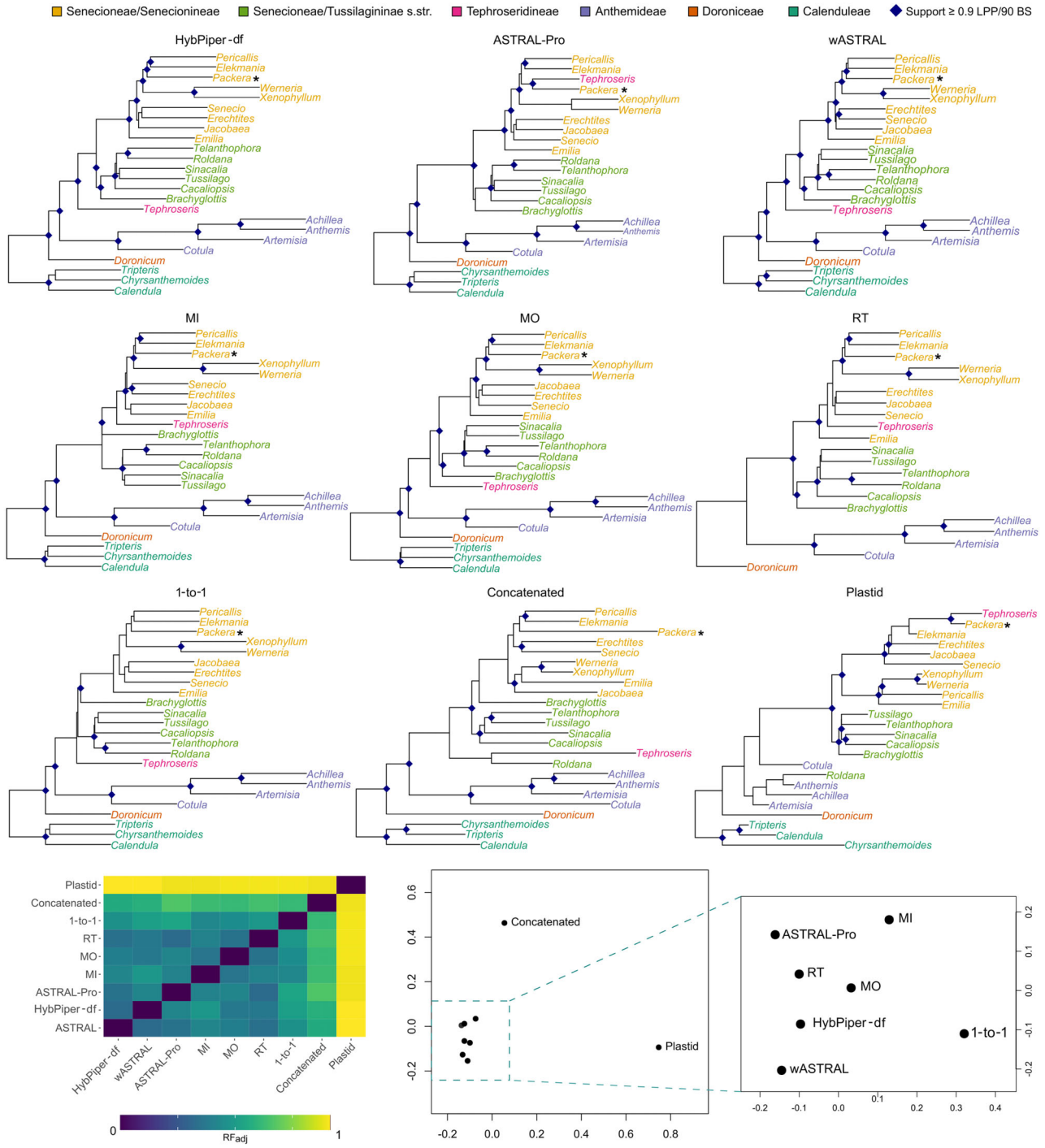


FIGURE 1. Genus-level phylogenies of all trees generated in this study. Genera are colored according to their tribal/subtribal classifications following the key on the top of the figure. *Packera* is marked with an asterisk next to its placement in the trees. Diamonds at some nodes represent support values over 0.9 LPP or 90 BS. Bottom left is a matrix of RF_{adj} values between the nine different trees, with most similar topologies as dark blue squares (0) and least similar as yellow squares (1). The bottom right shows the RF_{adj} values plotted as a principal component analysis (PCA). Points closer together indicate higher similarity, whereas dots further away indicate higher dissimilarity.

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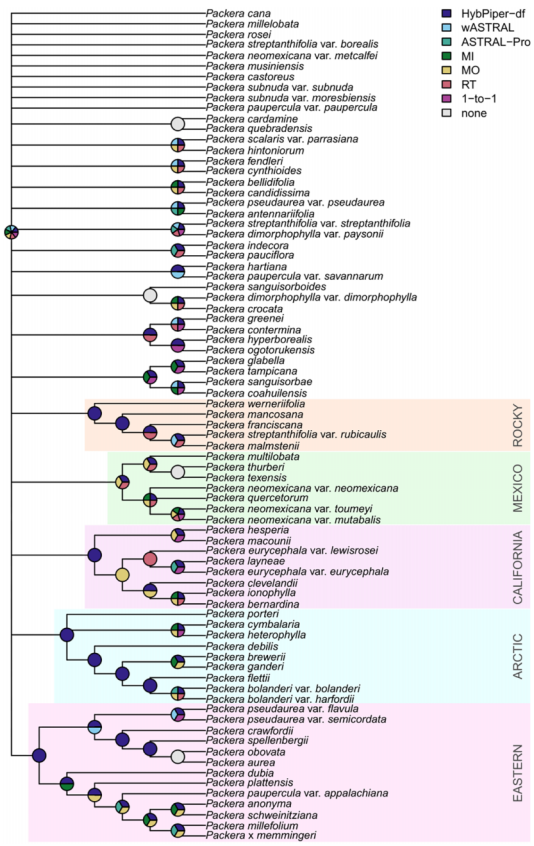


FIGURE 2. Majority-rule consensus tree comparing the node relationships of all *Packera* taxa across the seven ASTRAL-based methods used in this study. Pie charts at each node indicate the phylogenomic method that retains that node and relationship following the key on the top right. Highlighted clades indicate clades where reticulation events were tested.

concatenated tree also had the highest proportion of counter-support nodes (QC < 0: 26.4%; Table 1; see Supplementary Fig. S7 for all Quartet Sampling results). PhyParts, which quantifies how many gene trees support or conflict with each species tree branch, indicated that the HybPiper-df and MO trees had the most concordant gene trees. The wASTRAL analysis had the highest percentages of gene trees supporting the top alternative bipartition or second best topology. The MI tree harbored the highest discordance and uninformative data (Supplementary Fig. S8).

This underlying discordance could be caused by ILS, which was common in all tested ASTRAL trees (Table 1 ; Supplementary Fig. S9). Introgressive hybridization (IH) was not detected without the presence of ILS in any of the nodes, though IH was most common in the nodes that were shown to be discordant (Supplementary Figs. 7 and 8). Interestingly, crown *Packera* consistently had high IH values, potentially indicating introgression at that node. Backbone nodes were generally considered well resolved with high confidence (ILS index < 50% and IH index < 10%; Shang et al. 2025), especially in the outgroups, whereas *Packera* taxa appeared less resolved.

Molecular Clock Dating

The resulting divergence rate estimates differed among HybPiper-df, 1-to-1, and MO (Supplementary Table S11 and Supplementary Fig. S10), with estimated ages of the MO and 1-to-1 trees older and sometimes almost double the HybPiper-df ages. For HybPiper-df, the mean crown age for *Packera* ranged from 11.4 myr in scenarios 5 and 6 to 13.5 myr in scenarios 1–4. Alternatively, the mean crown age of *Packera* was almost doubled in the 1-to-1 and MO analyses, with ranges from ∼21 myr in scenario 5 to 35.5–35.7 myr in scenario 4 (Supplementary Table S11 and Supplementary Fig. S10). Scenario 2 was chosen because it incorporated all available primary calibration points and yielded divergence time estimates that align well with known patterns of evolutionary history (Pelser et al. 2010; Mandel et al. 2019), while minimizing reliance on secondary calibrations (Supplementary Table S6) (Müller and Reisz 2005; Benton and Donoghue 2007; Parham et al. 2012). The dated trees from all six scenarios can be found in Supplementary Figure S10.

Biogeography

Evaluation of the six different model choices, DEC, DIVALIKE, and BAYAREALIKE with and without jump dispersals (+J), for ancestral range evolution demonstrated that BAYAREALIKE was the best fit given our data based on its lowest log likelihood (Ln L) and highest AIC score, as well as the highest weighted AICc score for all three of our analyses (Supplementary Table S12). BAYAREALIKE is a likelihood implementation of the BayArea-like model (Landis et al. 2013), which models geographic range evolution without an explicit vicariance parameter, allowing widespread ancestors and sympatric range subdivision, but not founder-event speciation unless +J is added. In our analysis, models including +J provided a better fit than those without in the DEC and DIVALIKE models, and were the same or very similar in the BAYAREALIKE models (Supplementary Table S12), indicating a potential for jump dispersal or founder-event speciation in our data. Trees of all model runs are in Supplementary Figure S11.

The historical biogeography analysis revealed distinct geographic patterns across the different orthology methods (Fig. 4; Supplementary Fig. S11). HybPiper-df identified the ancestral range of crown *Packera* as concentrated in the North American Prairies Province, Rocky Mountain and Vancouverian Province, and Madrean regions. In contrast, MO showed the ancestral regions of crown *Packera* being only the Rocky Mountain and Vancouverian Province, and Madrean regions. 1-to-1 was intermediate with crown *Packera* being ancestral to the North American Prairies Province, Rocky

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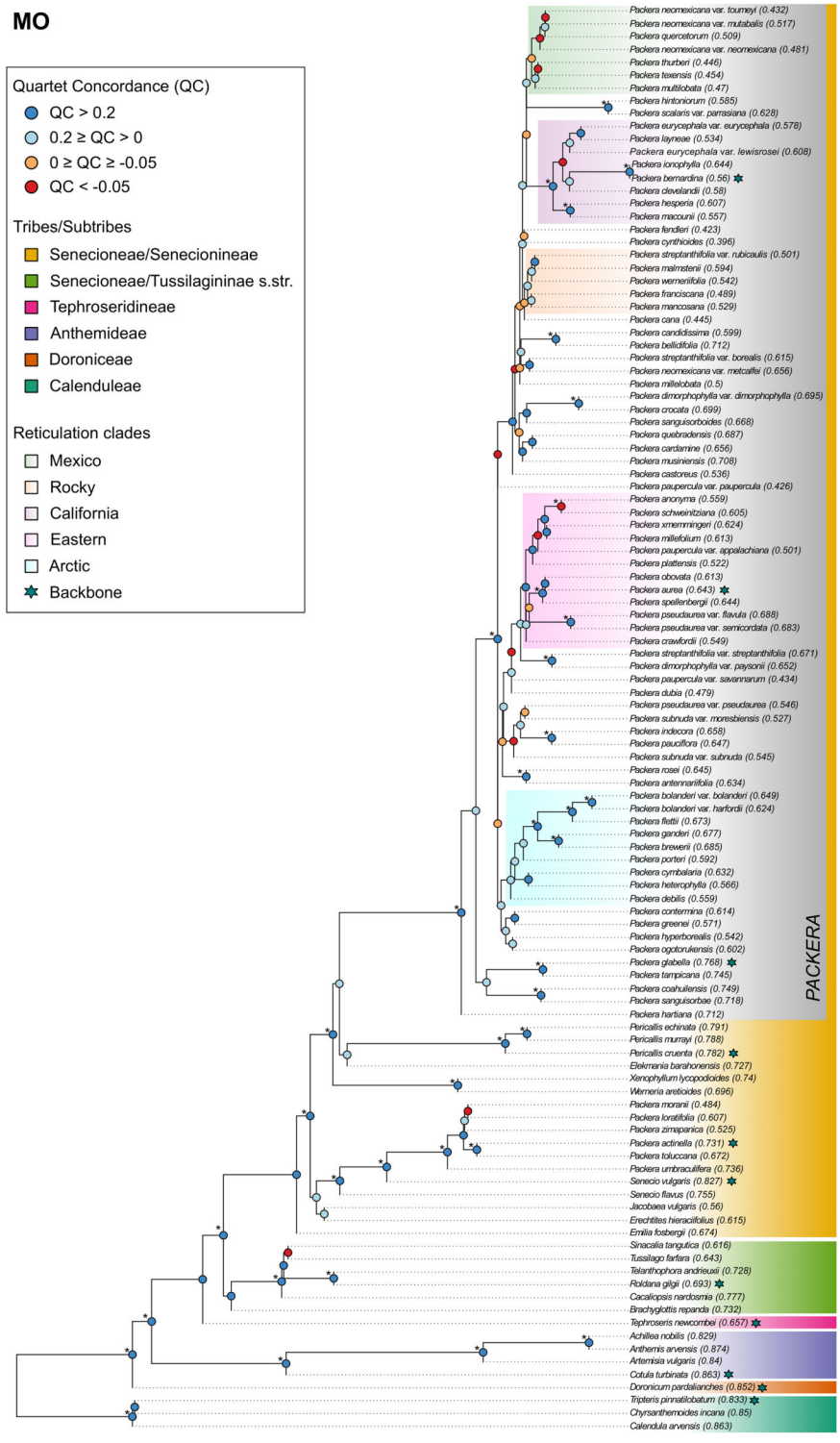


FIGURE 3. Full phylogeny of the "monophyletic outgroup" (MO) analysis. Tribal/subtribal status is indicated on the right of the phylogeny and follows the colors as indicated in the key on the left. Numbers next to tip labels correspond to the Quartet Fidelity (QF) score from Quartet Sampling, which ranges from 0 to 1, with values closer to 0 indicating erroneous taxa. Circles at nodes show the Quartet Concordance (QC) factors from Quartet Sampling, where red and orange indicate counter-support nodes. Asterisks next to QC circles indicate high support (≥95 LPP) at that node. Highlighted clades and stars next to some tip labels indicate clades where reticulation events were tested, following the color key.

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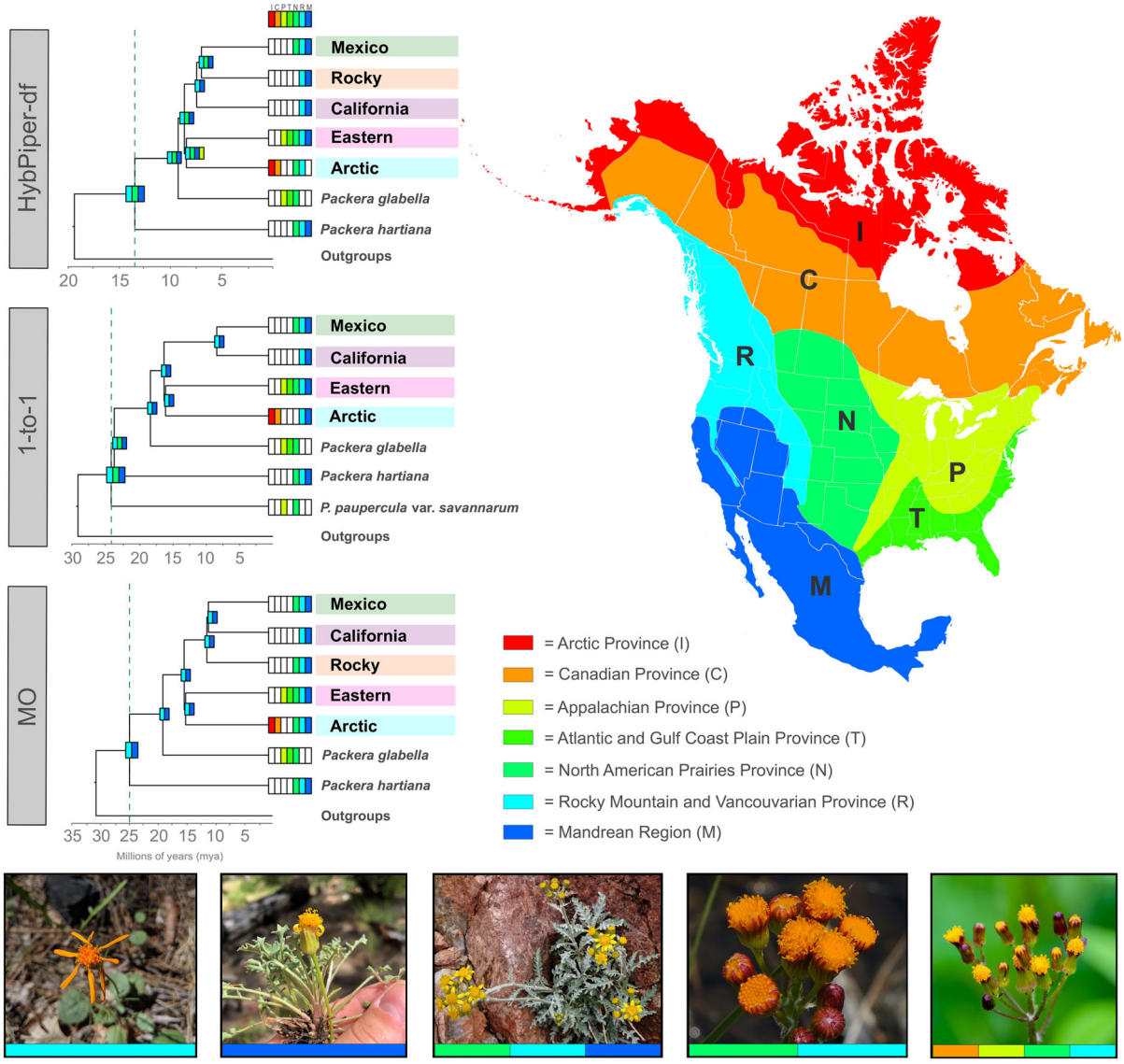


FIGURE 4. Predicted biogeographic region summaries generated from the BAYAREALIKE model in BioGeoBEARS of the HybPiper-df, 1-to-1, and MO analyses. The dated trees were generated using RelTime as input. Units at the bottom of the dated trees represent time in millions of years (myr). The map of North America is colored according to the geographic regions outlined by Takhtajan (1986): Arctic Province (I), Canadian Province (C), Appalachian Province (P), Atlantic and Gulf Coast Plain Province (T), North American Prairies Province (N), Rocky Mountain and Vancouvarian Province (R), and Madrean Region (M). Phylogenies for each analysis are simplified to the more major clades of *Packera*, with at least one taxon represented from each clade. Larger clades are designated by the reticulation clades identified in Figure 2 and Supplementary Figures 5 and 6. The bars to the left of the clade tip labels represent current distributions of the species. Bars at each node represent the predicted ancestral range at that node. All BioGeoBEARS results can be found in Supplementary Figure S11. Images of *Packera* species highlighting some of the morphological diversity seen in the group. Colored bars on the bottom of the image correspond to their current distributions, following the color scheme of the map and key. From left to right: Flame ragwort, *Packera greenei* (A.Gray) W.A.Weber & Á.Löve; Tehachapi ragwort, *P. ionophylla* (Greene) W.A.Weber & Á.Löve; Fendler's ragwort, *P. fendleri* (A.Gray) W.A.Weber & Á.Löve; Weak groundsel, *P. debilis* (Nutt.) W.A.Weber & Á.Löve; and Rayless mountain ragwort, *P. indecora* (Greene) Á.Löve & D.Löve. Image credits: Erika R. Moore-Pollard and Robert Lagier.

Mountain and Vancouverian Province, and Madrean regions, similar to HybPiper-df; however, the remaining ancestral range on the backbone was Rocky Mountain and Vancouverian Province and Madrean regions, similar to MO. Notably, all analyses showed one major expansion into the eastern United States—Appalachian Province and Atlantic and Gulf Coast Plain Province—represented by the Eastern clade (Fig. 4; Supplementary Fig. S11), with a few instances of individual introductions to the eastern United States. Many of these

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taxa originated initially in the North American Prairies Province and/or Rocky Mountain and Vancouverian Province ranges. HybPiper-df and MO showed two early expansions into the Arctic Province (I) and Canadian Province at the base of the phylogeny, whereas 1-to-1 showed at least three expansions north. 1-to-1 exhibited the most geographic inconsistency, followed by MO. Even so, larger clades consistently reflected broad geographic patterns across all analyses, although some widespread species did not always align with the geographic groupings consistently within geographic clades (Fig. 2; Supplementary Figs. 5, 6, and 11).

Reticulation Events

Five *Packera* clades were tested for ancient reticulation events given a large number of counter-support (QC < 0) nodes from QS on the HybPiper-df, 1-to-1, and MO trees (Fig. 2; Supplementary Figs. 5 and 6). Additionally, a single species from each major clade was then used to represent backbone relationships throughout the entire phylogeny, totaling six separate PhyloNetwork runs named: Arctic, Backbone, California, Eastern, Mexico, and Rocky (Supplementary Table S12). Log-likelihood values indicated that the most likely number of reticulation events (h) varied depending on the paralog processing method and clade (Supplementary Fig. S12). For example, no paralog processing method had a consistent h across clades. Alternatively, some clades had a consistent h independent of the paralog processing method used, particularly California, Rocky, and Eastern (Supplementary Table S8 and Supplementary Fig. S12). Arctic varied from h $= 2$ (1-to-1 and MO) to h $= 3$ (HybPiper-df). Backbone varied from h $= 2$ (HybPiper-df and MO) to h $= 6$ (1-to-1), which was the highest estimated number of reticulations inferred. Mexico varied from h $= 1$ (MO) to h $= 3$ (HybPiper-df and 1-to-1) (Fig. 5; Supplementary Fig. S12 and Supplementary Table S8). The taxa also differed between analyses given different phylogenetic relationships between clades (Supplementary Table S13), impacting some of the networks identified.

Some networks showed different events depending on the phylogenetic method and subset of data (Fig. 5); however, some consistent patterns emerged. For example, the Eastern clade continuously showed the same reticulation event regardless of the paralog processing method, and parentage percentages were very similar. In contrast, the Rocky clade did not show the same reticulation events, but the networks generally involved *Packera mancosana* Yeatts, B.Schneid. & Al Schneid. The California clade showed differing results across methods: both HybPiper-df and 1-to-1 analyses identified *P. bernardina* (Greene) W.A.Weber & Á.Löve as the product of reticulation with the same parentage, whereas MO instead inferred *P. bernardina* as one of the parental lineages contributing to *P. eurycephala* var. *lewisrosei* (J.T.Howell) J.F.Bain. Thus, differences among methods primarily reflected shifts in reticulation directionality rather than disagreement about which lineages were involved. A similar reversal was observed in the second inferred event, where HybPiper-df identified *P. macounii* (Greene) W.A.Weber & Á.Löve as a parent of *P. layneae* (Greene) W.A.Weber & Á.Löve, whereas MO recovered the opposite relationship. The Mexico clade was generally inconsistent, but featured reticulation events involving *P. neomexicana* (A.Gray) W.A.Weber & Á.Löve and its varieties. Additionally, *P. texensis* O'Kennon & Trock and *P. quercetorum* (Greene) C.Jeffrey were involved in reticulation events in two of the three paralog processing networks. Among all the clades, the Arctic clade exhibited the most variability in the number of reticulation events, with the *P. bolanderi* (A.Gray) W.A.Weber & Á.Löve varieties consistently involved in some form of reticulation. However, the specifics of these events varied depending on sampling and paralog processing methods. In the Backbone subtree analysis, *Tephroseris* was frequently identified as a product of reticulation, with 1-to-1 and MO showing the same parentage. In contrast, HybPiper-df did not show the same reticulation events but did identify a reticulation event involving *Tephroseris* with a different parentage. Notably, the parentage of *Tephroseris* always included a *Packera* taxon, though the specific taxon varied between two sister species, suggesting that differences among analyses may reflect gene sampling rather than fundamentally distinct hybrid origins.

Discussion

Gene duplication was pervasive in *Packera*, with more than 80% of nuclear loci showing evidence of paralogy (Supplementary Table S3). Such widespread paralogy poses major challenges for species tree inference, especially in Hyb-Seq datasets, where probes may unintentionally enrich for multiple homeologs (Salmon et al. 2012; Akama et al. 2014; Crameri et al. 2022), complicating the recovery of true single-copy orthologs for phylogenetic reconstructions. Although some paralogy is expected in target enrichment studies (Siniscalchi et al. 2021), the extent observed in *Packera* is unusually high and reflects lineage-specific genome dynamics, rather than a general feature of all species. This has important methodological implications, as HybPiper's orthology inference relies largely on single-copy filtering (Johnson et al. 2016), and using its outputs alone can introduce noise and bias into gene tree reconstruction, ultimately distorting species tree inference. Consistent with our study, reliance on the standard HybPiper orthology designations without additional paralog-aware filtering or modeling was insufficient for accurate phylogenetic inference in *Packera* (Supplementary Fig. S5), a group characterized by extensive polyploidy and hybridization (Trock 2006). These challenges are not unique to *Pack-*

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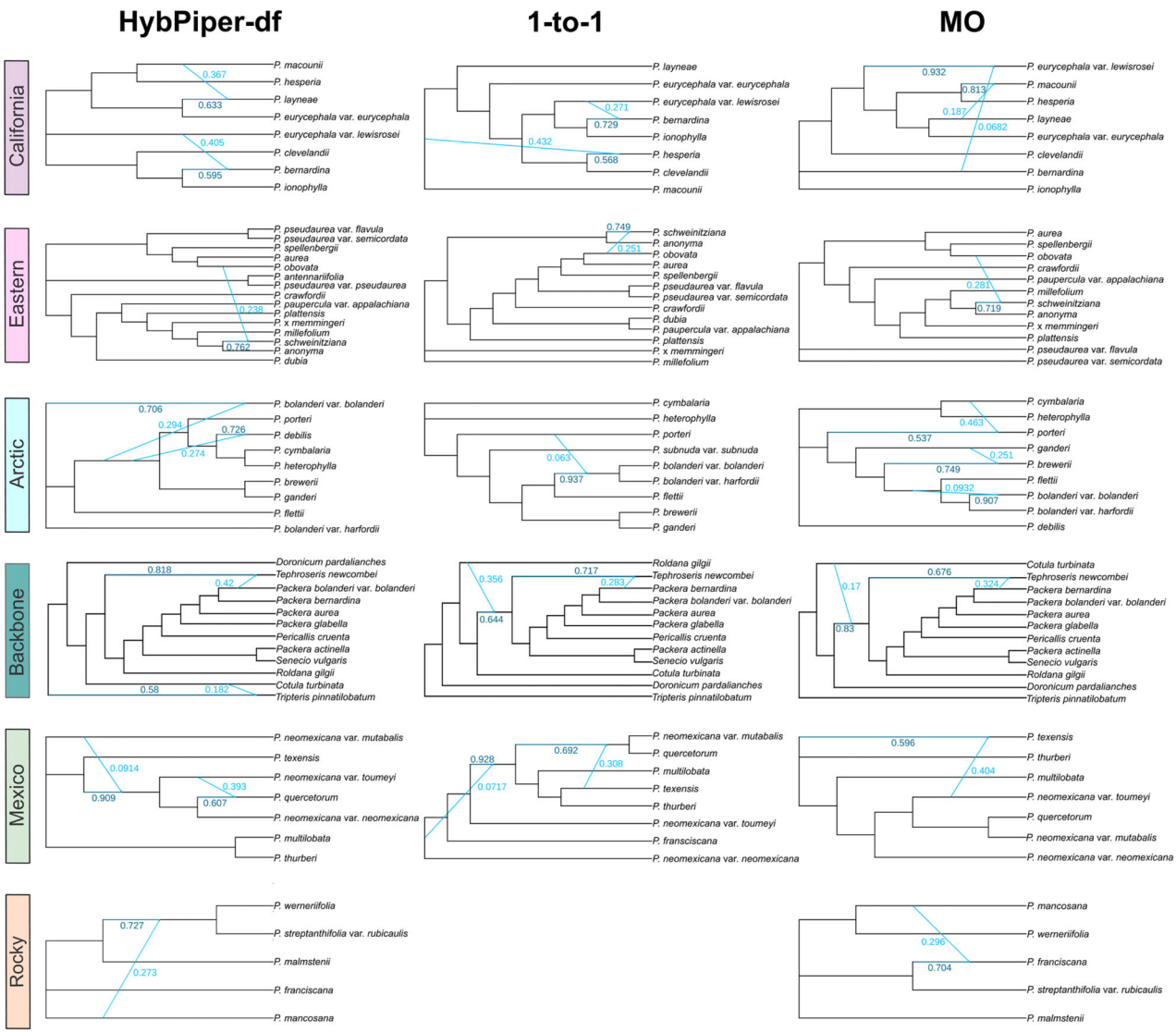


FIGURE 5. PhyloNetworks figures of the three tested analyses: HybPiper-df, 1-to-1, and MO. The various areas of the phylogeny that were tested are shown on the left and can be read as a row. Dark blue edges indicate the major hybrid edge, whereas the light blue shows the minor hybrid edge. The associated numbers next to each branch correspond to the percentage of parentage given by that branch. The number of hybridizations corresponds to the best log likelihood value. The 1-to-1 analysis does not have a Rocky analysis because the clade in that tree only had four taxa.

era, as many plant and non-plant lineages also experience difficulties reconstructing phylogenies due to hybridization and gene/genome duplications, highlighting the complexity of orthology inference in phylogenomics and the need for rigorous paralog processing across systems.

The most recent phylogeny of *Packera* prior to this study was based on ITS data and revealed extremely low sequence variation, with only 0–5% divergence and several *Packera* taxa sharing identical sequences (Bain and Golden 2000). This led to a largely unresolved tree, likely due to few phylogenetically informative sites, sparse species sampling, and potential reticulation events (Funk 1985; Linder and Rieseberg 2004). By using Hyb-Seq data and broader taxon sampling, we substantially improved phylogenetic resolution across *Packera*, including within major clades. However, our analyses also revealed that gene tree concordance and species tree resolution varied substantially across methods (Figs. 1 and 2; Table 1 ; Supplementary Fig. S4), exemplifying the influence of paralog processing strategies on downstream phylogenetic outcomes.

Methods that explicitly model gene duplication and loss, such as those implemented in paragone-nf (Yang and Smith 2014; Jackson et al. 2023), generally yielded more concordant gene trees and better-resolved species

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trees (Fig. 2; Supplementary Fig. S6). These approaches consistently showed higher overall support values and reduced topological instability compared with more aggressively filtered data sets. Although difficult to disentangle, these results suggest that accounting for duplication and loss may improve phylogenetic signal retention in this system. However, substantial conflict persisted across the trees (Figs. 1 and 2), indicating that even sophisticated modeling approaches cannot fully resolve deep relationships in groups with complex histories like *Packera*. These discrepancies may in part reflect differences in the number and identity of genes retained after filtering (Table 1; Supplementary Table S4). For instance, the 1-to-1 method, which excludes all loci with paralogs flagged by HybPiper and resembles approaches used in tools like Phyluce, eliminates a significant portion of the data set and may oversimplify the data, especially in duplication-rich systems such as *Packera*. Consistent with this reduction in gene sampling, the 1-to-1 data set generally produced lower overall support and greater topological instability relative to methods that retained paralog information, suggesting that aggressive orthology filtering may compromise phylogenetic resolution in this system.

These performance differences likely reflect not only gene sampling depth but also how orthology is inferred and sequences are assembled. Although Phyluce and 1-to-1 filtering both remove multi-copy genes, the underlying assembly and orthology inference methods differ (Herrando-Moraira et al. 2018), causing Phyluce to outperform HybPiper in certain cases (e.g., Mandel et al. 2019; Thapa et al. 2020). Similarly, the RT method naturally loses information because it removes outgroups to determine orthology (Yang and Smith 2014; Morales-Briones, Gehrke, et al. 2021; Morales-Briones, Kadereit, et al. 2021) and potentially limits phylogenetic signal, contributing to the instability. Together, these comparisons reinforce that methodological differences in assembly, orthology inference, and gene filtering can substantially influence phylogenetic outcomes. Although some approaches improved concordance in *Packera*, none fully resolved the underlying discordance, emphasizing that method choice must be guided by both biological context and data set characteristics.

Results from previous phylogenetic work suggest that evolutionary relationships in *Packera* tend to loosely follow a geographic pattern (Bain and Golden 2000), which is largely consistent with our findings (Fig. 2; Supplementary Figs. 5, 6, and 11). Our nuclear phylogenies showed that major clades tended to contain taxa within similar geographic ranges, independent of which paralog processing method was utilized. Additionally, most taxa within these major clades were conserved across methods, though their placement along the backbone, and some taxa, did vary (Figs. 2 and 3; Supplementary Figs. 5 and 6). For example, the Mexico clade was the most consistently recovered multi-taxon grouping across analyses, indicating that at least some regional structure is robust to methodological differences (Fig. 2). In contrast, the Eastern clade showed greater topological instability despite containing a largely overlapping set of taxa across orthology inference methods. Several widespread taxa shifted in and out of this clade among analyses, contributing to backbone uncertainty (Fig. 2; Supplementary Fig. S4). However, conflicting signals may also reflect unrecognized lineage structure within widespread taxa or the effects of incomplete sampling, both of which can complicate phylogenetic inference even in the absence of hybridization.

Our results also reveal that evolutionary interpretations such as divergence dating and biogeographic reconstruction are highly sensitive to the paralog processing approach used to infer the underlying phylogeny. Divergence time estimates varied across methods, with some species trees suggesting that crown *Packera*'s diversification occurred more recently (HybPiper-df: 13.5 myr), whereas others placed key divergences deeper in time (1-to-1: 24.2 myr and MO: 25 myr; Supplementary Fig. S10 and Supplementary Table S11). These differences in divergence times are not trivial, as they affect our understanding of how *Packera* taxa respond to climatic shifts, mountain uplift, and other historical events, and have broader implications for comparative studies across Asteraceae and other radiating plant lineages. For example, tribal- and family-wide studies predicted the crown age of *Packera* to be fairly young, between 7.8 and 20.2 myr (Pelser et al. 2010; Mandel et al. 2019; Zhang et al. 2021), which is more in-line with the HybPiper-df analysis. However, more *Packera*-specific predictions are older, with Barkley (1988) predicting the origin of *Packera* in the later stages of the mid-Tertiary (~22–37 myr), closely aligning with the paragone-nf analyses (Supplementary Fig. S10). Similarly, biogeographic patterns shifted depending on the inferred topology, with each analysis showing different migration patterns over time (Fig. 4,; Supplementary Fig. S11). Even so, the broad pattern is relatively unchanged. For example, all topologies generally agreed that *Packera* originated in western North America with one major expansion east and then later expansions north from the western clades. From there, individual taxa expanded or became specialized to certain regions, with most of the diversity present in western North America. Because the topologies are consistent across methods, we can be more confident that they likely reflect the true evolutionary history of *Packera* instead of methodological artifacts.

Despite improvements from paralogy-aware filtering, the species trees for *Packera* were not fully resolved, indicating that underlying biological processes such as ILS, hybridization, and introgression also play significant roles in shaping *Packera*'s evolutionary history. This is supported by strong cytonuclear discordance, where plastid phylogenetic relationships vary significantly from the nuclear trees (Fig. 1; Supplementary Fig. S4), consistent with chloroplast capture or ancient

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hybridization events (Doyle 1992; Maddison 1997; Galtier and Daubin 2008; Soltis and Soltis 2009; Smith et al. 2015; Stull et al. 2020).

Directly testing for gene flow revealed multiple introgression events, particularly along short internal branches susceptible to ILS (Supplementary Fig. S9). However, the specific reticulation patterns inferred were sensitive to how paralogs were handled, with alternative processing strategies yielding different hybridization scenarios. For example, the Mexico clade provides the clearest example of this instability. Although overall clade membership was relatively consistent across paralog processing strategies (Fig. 2), relationships within the clade were weakly supported and highly discordant in all species tree analyses (Fig. 2; Supplementary Figs. 4–7). Network inferences further amplified this uncertainty: reticulation placements varied among analyses, and no single hybridization scenario was consistently recovered. This complexity mirrors the clade's documented taxonomic and evolutionary history. Species such as *P. neomexicana* have long been recognized as morphologically variable and broadly distributed, with suggested affinities to *P. quercetorum* and *P. multilobata* (Torr. & A. Gray) W.A. Weber & Á. Löve based on morphology and earlier molecular work (Bain and Jansen 1995; Trock 1999). Likewise, *P. texensis* has been described as resembling *P. thurberi* (A. Gray) B. L. Turner and appears closely allied in several of our phylogenies, yet its placement shifts across analyses. The combination of morphological overlap, geographic proximity, and prior hypotheses of shared evolutionary history provides a plausible biological explanation for the method-dependent reticulation patterns observed here.

In contrast, the Eastern clade exhibited greater consistency in inferred hybridization despite historical expectations of widespread gene flow. Taxonomic treatments have long suggested extensive interspecific hybridization among eastern North American *Packera*, particularly involving *P. pseudaurea* (Rydb.) W.A.Weber & Á.Löve, which has been considered taxonomically problematic due to suspected hybrid origins (Barkley 1962; Trock 1999). Surprisingly, none of our network analyses detected introgression involving *P. pseudaurea*. Instead, all paralog processing strategies consistently recovered a reticulation between *P. obovata* (Willd.) W.A.Weber & Á.Löve and *P. schweinitziana* (Nutt.) W.A.Weber & Á.Löve. Although this species pair has not been emphasized in previous taxonomic treatments and lacks formal crossing studies (Kowal 1975), their overlapping ecological preferences and broad geographic ranges provide opportunities for gene flow. The repeated recovery of this reticulation across data sets suggests that this signal is robust to orthology inference strategy.

Together, these results demonstrate a critical point: gene selection and paralog processing directly shapes reticulation inference. Because each gene has its own evolutionary trajectory, shaped by recombination and gene duplication and loss (Betrán 2015), decisions about orthology filtering can either obscure or reveal historical hybridization. In clades with complex histories, such as in the Mexico clade, alternative gene selection strategies can lead to substantially different reticulation hypotheses. In contrast, strongly supported introgression signals, such as the reticulation between *P. obovata* and *P. schweinitziana* in the Eastern clade, may persist across analytical treatments. These findings underscore the importance of explicitly evaluating how paralog handling influences evolutionary network reconstruction.

Network analyses, such as PhyloNetworks, are inherently sensitive to sampling bias (Solís-Lemus and Ané 2016). Even though we sampled 85 out of a targeted 91 *Packera* taxa (~93% coverage of the genus), missing taxa may have impacted our results. Including the remaining and unsampled *Packera* taxa in future studies could help increase our understanding of these relationships. Additionally, limiting the analyses to specified clades can ignore other potential reticulation events occurring across groups. We recognize this limitation; however, expanding sampling to more taxa or loci is currently restricted by the intensive and often unfeasible computational requirements of reticulation network analyses (see Hejase and Liu 2016). With recent bioinformatic advances and increased interest in non-bifurcating trees, we anticipate that less computationally intense and more efficient phylogenetic and network tools will soon be developed, allowing researchers to test for reticulations across large phylogenies.

Ghost introgressions, which are ancient introgression events that leave genetic signatures of extinct species in present-day species (Ottenburghs 2020), also pose challenges to inferences made by hybridization networks. Accounting for ghost introgressions in macroevolutionary studies may help improve the accuracy of trait evolution and biogeographic reconstructions, refine divergence time estimates, and reveal hidden contributions of hybridization to diversification (Ottenburghs 2020; Tricou et al. 2022). Network methods such as PhyloNetworks can detect ghost introgressions, though they do not specify which hybrid edge is from a ghost introgression event, so it is up to the researcher to recognize its presence, which can be difficult especially without background knowledge of the group (Solís-Lemus and Ané 2016; Solís-Lemus et al. 2017).

Our findings emphasize that accurately reconstructing reticulate evolutionary histories requires embracing the independent histories of genes and carefully considering how orthology is defined. In *Packera*, gene tree heterogeneity is expected and evolutionarily informative rather than mere noise. Methods that oversimplify gene relationships, such as pipelines restricted to single-copy orthologs, risk masking the very patterns needed to understand complex evolutionary processes. By comparing alternative strategies for handling paralogs and in-

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tegrating gene flow analyses, we show that approaches that explicitly model gene duplication and loss provide modest improvements in resolution and concordance but are insufficient on their own to fully resolve relationships. Direct tests for reticulation and ILS revealed the extent to which *Packera*'s evolutionary history has been shaped by hybridization and gene flow, underscoring the broader challenge of capturing complexity in phylogenomic data sets. Within polyploid systems, the tradeoff between excluding paralogs to reduce conflict and retaining them to preserve evolutionary signals becomes especially pronounced.

These results also highlight how paralog handling influences not only phylogenetic reconstruction but also the biological inferences drawn from the inferred relationships. Macroevolutionary analyses such as divergence dating, trait evolution, and biogeography are highly sensitive to orthology inference strategies and are only as reliable as the phylogenetic framework on which they are built (e.g., Magalhaes et al. 2021). Improving that framework will require both methodological innovation and deeper biological insight. For example, targeted probe sets designed to enrich truly single-copy loci (e.g., Compositae-ParaLoss-1272; Moore-Pollard et al. 2024) offer one path forward. Yet entirely excluding paralogs may discard informative loci, while retaining them without appropriate modeling risks misleading inference. Integrating morphological, ecological, and geographic evidence is therefore essential for assessing whether patterns inferred from large-scale data sets reflect genuine evolutionary history or methodological artifacts.

Rather than treating paralogy as a technical obstacle to be circumvented in phylogenetic studies, our findings underscore its potential to illuminate fundamental evolutionary processes, including gene duplication, polyploidy, and reticulation, that have shaped the diversification of lineages across the tree of life. Although these challenges are common to many plant and nonplant groups, our data set is uniquely suited to tackle this issue given dense taxon sampling and high locus recovery across both orthologs and paralogs, providing a uniquely powerful system for testing and improving phylogenomic methods. Additionally, having decades of natural history and systematics work in the group enabled us to conduct more thorough investigations into these challenges. As the field advances toward increasingly nuanced and biologically realistic models of genome evolution, systems like *Packera* will be essential for developing tools that can fully capture the intricacy of complex evolutionary histories.

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Supplementary Data

Supplementary material is available at *Systematic Biology* online.

Conflict of Interest

The authors declare no conflict of interest.

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Data Availability

Raw sequence data are available in the National Center for Biotechnology Information (NCBI) Sequence Read Archive (SRA) under BioProjects PRJNA907383, PRJNA978568, PRJNA540287, and PRJNA516161.

Data available from the Dryad Digital Repository: https://dx.doi.org/10.5061/dryad.bg79cnppn.

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